

Vercel Environment Variables Migration Guide

Overview

This guide provides step-by-step instructions for migrating all 34 environment variables from your local `.env` file to Vercel's deployment platform.

Last Updated: 2025-10-19

Total Variables: 34 (across 7 categories)

Deployment Platform: Vercel

Quick Reference

Variable Categories

1. **Database Configuration** (1 variable)
 2. **NextAuth Configuration** (2 variables)
 3. **Stripe API Keys** (3 variables)
 4. **Stripe Product & Price IDs** (12 variables)
 5. **Blockchain & Smart Contract** (4 variables)
 6. **NFT.Storage/IPFS** (1 variable)
 7. **Cloudflare** (6 variables)
 8. **Application Configuration** (2 variables)
 9. **Feature Flags** (4 variables)
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Migration Methods

Method 1: Vercel Dashboard (Recommended for First-Time Setup)

Step 1: Access Environment Variables

1. Log in to your Vercel dashboard at <https://vercel.com>
2. Select your project (or create a new one)
3. Navigate to **Settings** → **Environment Variables**

Step 2: Add Variables by Category

Copy and paste each variable name and value from the sections below.

Environment Variables by Category

1. Database Configuration

Purpose: PostgreSQL database connection for production

Variable Name	Example Value	Production Value	Required
DATABASE_URL	postgresql:// user:pass@host:5432/ db	Use your production DB URL	✔ Yes

Production Note:

✔ **Already configured with managed PostgreSQL** at db-aa342a793.db001.hosteddb.reai.io

⚠ Ensure this database is accessible from Vercel's deployment regions

```
# Format
DATABASE_URL="postgresql://[username]:[password]@[host]:[port]/[database]?con-
nect_timeout=15"
```

Vercel CLI Command:

```
vercel env add DATABASE_URL production
```

2. NextAuth Configuration

Purpose: Authentication and session management

Variable Name	Production Value	Required	Notes
NEXTAUTH_SECRET	Generate new for production	✔ Yes	Must be unique and secure
NEXTAUTH_URL	Your Vercel deploy- ment URL	✔ Yes	Update after first de- ployment

Generate NEXTAUTH_SECRET:

```
openssl rand -base64 32
```

Initial Setup:

```
# For first deployment, use temporary URL
NEXTAUTH_URL="https://your-project.vercel.app"

# After deployment, update with actual URL
NEXTAUTH_URL="https://authichain.vercel.app"
```

Vercel CLI Commands:

```
vercel env add NEXTAUTH_SECRET production
vercel env add NEXTAUTH_URL production
```

3. Stripe API Keys

Purpose: Payment processing (currently TEST MODE)

⚠ IMPORTANT: Current keys are TEST MODE. Update with LIVE keys for production.

Variable Name	Current Mode	Production Required
STRIPE_SECRET_KEY	Test Mode	✓ Live Key Required
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY	Test Mode	✓ Live Key Required
STRIPE_WEBHOOK_SECRET	Test Mode	✓ Live Key Required

Get Production Keys:

1. Log in to Stripe Dashboard: <https://dashboard.stripe.com>
2. Toggle from **Test Mode** to **Live Mode** (top right)
3. Navigate to **Developers** → **API Keys**
4. Copy **Secret key** and **Publishable key**

Webhook Setup:

1. Go to **Developers** → **Webhooks**
2. Add endpoint: `https://your-project.vercel.app/api/webhooks/stripe`
3. Select events: `checkout.session.completed`, `customer.subscription.updated`, `customer.subscription.deleted`, `invoice.payment_succeeded`, `invoice.payment_failed`
4. Copy the **Signing secret** (starts with `whsec_`)

Vercel CLI Commands:

```
vercel env add STRIPE_SECRET_KEY production
vercel env add NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY production
vercel env add STRIPE_WEBHOOK_SECRET production
```

4. Stripe Product & Price IDs

Purpose: Subscription tiers and pricing

⚠ IMPORTANT: Some price IDs show placeholders (`price_xxxxxxxxxxxx`). Update with actual production IDs.

Variable Name	Tier	Billing Cycle	Status
STRIPE_CREATOR_PRICE_ID	Creator	Default	✓ Configured
STRIPE_CREATOR_MONTHLY_PRICE_ID	Creator	Monthly	✓ Configured
STRIPE_CREATOR_ANNUAL_PRICE_ID	Creator	Annual	⚠ Needs Update
STRIPE_PRO_PRICE_ID	Pro	Default	✓ Configured
STRIPE_PRO_MONTHLY_PRICE_ID	Pro	Monthly	✓ Configured
STRIPE_PRO_ANNUAL_PRICE_ID	Pro	Annual	⚠ Needs Update
STRIPE_ENTERPRISE_PRICE_ID	Enterprise	Default	⚠ Needs Update
STRIPE_ENTERPRISE_MONTHLY_PRICE_ID	Enterprise	Monthly	⚠ Needs Update
STRIPE_ENTERPRISE_ANNUAL_PRICE_ID	Enterprise	Annual	⚠ Needs Update
STRIPE_AGENCY_PRICE_ID	Agency	Default	⚠ Needs Update
STRIPE_AGENCY_MONTHLY_PRICE_ID	Agency	Monthly	⚠ Needs Update
STRIPE_AGENCY_ANNUAL_PRICE_ID	Agency	Annual	⚠ Needs Update
STRIPE_BRAND_PRICE_ID	Brand	Default	✓ Configured

Get Production Price IDs:

1. In Stripe Dashboard, go to **Products**
2. Select each product (Creator, Pro, Enterprise, Agency, Brand)
3. Under **Pricing**, find the price ID for each billing cycle
4. Format: `price_1234567890abcdef`

Batch Add via Vercel CLI:

```
# Creator Tier
vercel env add STRIPE_CREATOR_PRICE_ID production
vercel env add STRIPE_CREATOR_MONTHLY_PRICE_ID production
vercel env add STRIPE_CREATOR_ANNUAL_PRICE_ID production

# Pro Tier
vercel env add STRIPE_PRO_PRICE_ID production
vercel env add STRIPE_PRO_MONTHLY_PRICE_ID production
vercel env add STRIPE_PRO_ANNUAL_PRICE_ID production

# Enterprise Tier
vercel env add STRIPE_ENTERPRISE_PRICE_ID production
vercel env add STRIPE_ENTERPRISE_MONTHLY_PRICE_ID production
vercel env add STRIPE_ENTERPRISE_ANNUAL_PRICE_ID production

# Agency Tier
vercel env add STRIPE_AGENCY_PRICE_ID production
vercel env add STRIPE_AGENCY_MONTHLY_PRICE_ID production
vercel env add STRIPE_AGENCY_ANNUAL_PRICE_ID production

# Brand Tier
vercel env add STRIPE_BRAND_PRICE_ID production
```

5. Blockchain & Smart Contract Configuration

Purpose: Web3 wallet authentication and NFT minting

⚠ SECURITY WARNING: `PRIVATE_KEY` contains sensitive data

Variable Name	Purpose	Security Level	Required
<code>PRIVATE_KEY</code>	Contract deployment/ transactions	🔴 Critical	✅ Yes
<code>POLYGON_AMOY_RPC_URL</code>	Polygon testnet RPC	🟡 Medium	✅ Yes
<code>CONTRACT_ADDRESS</code>	Deployed NFT contract	🟢 Low	✅ Yes
<code>POLYGONSCAN_API_KEY</code>	Blockchain explorer API	🟡 Medium	✅ Yes

Production Recommendations:

1. `PRIVATE_KEY`:

- ⚠ **NEVER commit to git**
- Use a dedicated wallet for production (not your personal wallet)
- Fund with minimal MATIC for gas fees
- Store in Vercel as **sensitive** variable

2. `POLYGON_AMOY_RPC_URL`:

- Current: Testnet (Polygon Amoy)

- Production: Consider upgrading to Polygon Mainnet
- Alternative: Use Infura or Alchemy for better reliability
- Format: `https://polygon-mainnet.infura.io/v3/YOUR-PROJECT-ID`

3. **CONTRACT_ADDRESS:**

- Current testnet address: `0x4da4D2675e52374639C9c954f4f653887A9972BE`
- Deploy to mainnet for production
- Update after mainnet deployment

Vercel CLI Commands:

```
vercel env add PRIVATE_KEY production
vercel env add POLYGON_AMOY_RPC_URL production
vercel env add CONTRACT_ADDRESS production
vercel env add POLYGONSCAN_API_KEY production
```

6. NFT.Storage/IPFS Configuration

Purpose: Decentralized storage for NFT metadata and images

Variable Name	Status	Required	Notes
<code>NFT_STORAGE_API_KEY</code>	✅ Configured	✅ Yes	Production-ready

Production Note:

- ✅ API key is already configured and production-ready
- 📦 Free tier: 31 GB storage, unlimited uploads
- 🔄 Consider upgrading if expecting high volume

Vercel CLI Command:

```
vercel env add NFT_STORAGE_API_KEY production
```

7. Cloudflare Configuration

Purpose: CDN, caching, and R2 object storage

Variable Name	Purpose	Required	Notes
CLOUDFLARE_API_TOKEN	API authentication	✓ Yes	Zone-level access
CLOUDFLARE_ACCOUNT_ID	Account identifier	✓ Yes	Production-ready
R2_API_TOKEN	R2 storage auth	✓ Yes	Object storage
R2_ACCESS_KEY_ID	R2 access key	✓ Yes	Production-ready
R2_SECRET_ACCESS_KEY	R2 secret key	● Critical	Sensitive
R2_ENDPOINT	R2 bucket endpoint	✓ Yes	Production-ready

Production Note:

- ✓ All Cloudflare credentials are configured
- 📦 R2 Storage is production-ready alternative to S3
- 💰 More cost-effective than AWS S3 for high-traffic sites

Vercel CLI Commands:

```
vercel env add CLOUDFLARE_API_TOKEN production
vercel env add CLOUDFLARE_ACCOUNT_ID production
vercel env add R2_API_TOKEN production
vercel env add R2_ACCESS_KEY_ID production
vercel env add R2_SECRET_ACCESS_KEY production
vercel env add R2_ENDPOINT production
```

8. Application Configuration

Purpose: Core app settings

Variable Name	Development Value	Production Value	Required
NEXT_PUBLIC_SITE_URL	http://localhost:3000	Your Vercel URL	✓ Yes
NODE_ENV	development	production	✓ Auto-set

Setup Instructions:

1. NEXT_PUBLIC_SITE_URL:

- First deployment: Use temporary Vercel URL
- After deployment: Update with actual URL
- Custom domain: Update when domain is connected
- Format: `https://authichain.vercel.app` (no trailing slash)

2. NODE_ENV:

- Vercel automatically sets this to `production`
- No manual configuration needed

Vercel CLI Command:

```
vercel env add NEXT_PUBLIC_SITE_URL production
# NODE_ENV is auto-set by Vercel
```

9. Feature Flags

Purpose: Enable/disable features in production

Variable Name	Feature	Current Status	Recommended
<code>FEATURE_WALLET_AUTH</code>	Web3 Wallet Authentication	✓ Enabled	✓ Enable
<code>FEATURE_CRYPTOPAYMENTS</code>	Cryptocurrency Payments	✓ Enabled	✓ Enable
<code>FEATURE_AUCTIONS</code>	NFT Auctions	✓ Enabled	✓ Enable
<code>FEATURE_COLLECTIONS</code>	NFT Collections	✓ Enabled	✓ Enable

Production Recommendations:

- All features are production-ready
- Keep all enabled for full functionality
- Can disable individually if needed

Vercel CLI Commands:

```
vercel env add FEATURE_WALLET_AUTH production
vercel env add FEATURE_CRYPTOPAYMENTS production
vercel env add FEATURE_AUCTIONS production
vercel env add FEATURE_COLLECTIONS production
```

Method 2: Vercel CLI (Batch Import)

Prerequisites

```
# Install Vercel CLI
npm install -g vercel

# Login to Vercel
vercel login
```


Option A: Interactive Import (Recommended)

```
# Navigate to project
cd /home/ubuntu/authichain_premium/app

# Pull existing environment variables (if any)
vercel env pull

# Add all variables interactively
vercel env add DATABASE_URL production
vercel env add NEXTAUTH_SECRET production
# ... (continue for all 34 variables)
```

Option B: Bulk Import from File

Step 1: Create Production .env File

```
# Create a temporary production env file
cp .env .env.production.vercel
```

Step 2: Update Production Values

Edit `.env.production.vercel` and update:

- `NEXTAUTH_URL` → Your Vercel URL
- `NEXT_PUBLIC_SITE_URL` → Your Vercel URL
- Stripe keys → Live mode keys
- Price IDs → Production price IDs
- `NODE_ENV` → `production`

Step 3: Import via Vercel CLI

```
# Note: Vercel CLI doesn't have direct bulk import
# You'll need to add each variable manually
while IFS= read -r line; do
  if [[ $line =~ ^[A-Z_]+= ]]; then
    key="${line%*=}"
    value="${line#*=}"
    echo "Adding $key..."
    echo "$value" | vercel env add "$key" production
  fi
done < .env.production.vercel

# Delete temporary file after import
rm .env.production.vercel
```



Method 3: Vercel Dashboard (Bulk Paste)

Step 1: Prepare Variables

Create a text file with all variables in this format:

```
VARIABLE_NAME=value
ANOTHER_VARIABLE=another_value
```

Step 2: Paste in Dashboard

1. Go to Vercel Dashboard → Project Settings → Environment Variables
2. Click **“Add New”**
3. Paste all variables at once
4. Vercel will parse and add them automatically

Verification Checklist

Before Deployment

- ☐ All 34 environment variables added to Vercel
- ☐ Database URL is accessible from Vercel
- ☐ NEXTAUTH_SECRET is unique and secure (not development key)
- ☐ Stripe keys are in LIVE mode (not test mode)
- ☐ All Stripe price IDs are valid and production-ready
- ☐ PRIVATE_KEY is for a dedicated production wallet
- ☐ Webhook URLs match Vercel deployment URL
- ☐ Feature flags are set as desired

After First Deployment

- ☐ Update `NEXTAUTH_URL` with actual Vercel URL
- ☐ Update `NEXT_PUBLIC_SITE_URL` with actual Vercel URL
- ☐ Configure Stripe webhook with deployed URL
- ☐ Test all payment flows
- ☐ Test wallet authentication
- ☐ Test NFT minting and IPFS upload
- ☐ Verify database connections

Custom Domain (Optional)

- ☐ Connect custom domain in Vercel
- ☐ Update `NEXTAUTH_URL` with custom domain
- ☐ Update `NEXT_PUBLIC_SITE_URL` with custom domain
- ☐ Update Stripe webhook URL
- ☐ Update OAuth callback URLs (if applicable)

Security Best Practices

Critical Security Rules

1. Never Commit Sensitive Data

```
bash
# Always verify .gitignore includes:
.env
.env.*
!.env.example
```

2. Use Different Keys for Staging/Production

- Development: Test keys
- Staging: Test keys with production-like data
- Production: Live keys only

3. Rotate Keys Regularly

- Database passwords: Every 90 days
- API keys: Every 180 days
- NEXTAUTH_SECRET: Every 180 days
- Stripe webhooks: After any compromise

4. Limit Key Permissions

- Use read-only keys where possible
- Cloudflare: Zone-scoped tokens only
- Stripe: Restrict to required permissions

5. Monitor Key Usage

- Enable alerts for unusual activity
- Review access logs monthly
- Set up Vercel deployment protection



Troubleshooting

Issue: “Missing Environment Variable” Error

Solution:

```
# Verify all variables are set
vercel env ls

# Pull latest environment variables
vercel env pull

# Re-deploy
vercel --prod
```

Issue: Database Connection Fails

Cause: Vercel can't reach database host**Solution:**

1. Check database firewall rules
2. Allow Vercel's IP ranges
3. Use connection pooling (Prisma recommended)
4. Add `?connect_timeout=15` to DATABASE_URL

Issue: Stripe Webhook Failures

Cause: Incorrect webhook URL or secret**Solution:**

1. Update webhook URL in Stripe Dashboard
2. Regenerate webhook secret

3. Update `STRIPE_WEBHOOK_SECRET` in Vercel

4. Redeploy application

Issue: IPFS Upload Fails

Cause: Invalid NFT.Storage API key

Solution:

1. Verify key at <https://nft.storage>

2. Check API quota/limits

3. Regenerate key if expired

4. Update `NFT_STORAGE_API_KEY` in Vercel



Environment Variable Summary

By Security Level

Level	Count	Variables
● Critical	3	<code>PRIVATE_KEY</code> , <code>R2_SECRET_ACCESS_KEY</code> , <code>STRIPE_SECRET_KEY</code>
● Medium	11	Database, Auth, API keys
● Low	20	Public keys, URLs, flags

By Update Priority

Priority	Count	Action Required
⚠ High	15	Update before production
⚡ Medium	2	Update after first deploy
✓ Ready	17	No changes needed



Next Steps

1. Complete Environment Setup (Current Step)

- Add all 34 variables to Vercel
- Verify with checklist above

2. Initial Deployment

- Run `vercel --prod`
- Note the deployment URL

3. Post-Deployment Configuration

- Update URL-dependent variables
- Configure webhooks
- Test all features

4. Production Hardening

- Enable Vercel deployment protection
- Set up monitoring and alerts
- Configure custom domain (optional)

Support Resources

- **Vercel Documentation:** <https://vercel.com/docs/environment-variables>
- **Stripe Webhooks:** <https://stripe.com/docs/webhooks>
- **NFT.Storage:** <https://nft.storage/docs>
- **Cloudflare R2:** <https://developers.cloudflare.com/r2>



Maintenance Schedule

Weekly

- ☐ Monitor error logs in Vercel
- ☐ Check Stripe webhook deliveries
- ☐ Verify IPFS uploads

Monthly

- ☐ Review Vercel usage and costs
- ☐ Check R2 storage usage
- ☐ Audit environment variable access

Quarterly

- ☐ Rotate sensitive keys
- ☐ Update dependencies
- ☐ Review security best practices

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